



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 315: Microprocessor & Embedded System

Batch: 25th

Date: 09/12/2025

Mid-Term Examination

Semester: Fall 2025

Time: 1 Hour 30 minutes

Total Marks: 30

Answer any five questions of the following.

5 × 6 = 30

1. (a) Write difference between Microprocessor and Microcontroller. Explain the memory segmentation of 8086 Microprocessor? CLO3 3
(b) 8086 fetches several bytes ahead of execution and stores them in a queue. Explain how the execution unit interacts with this queue, and describe two situations when the prefetch queue is flushed or partially emptied, affecting performance. CLO3 3
2. (a) One part of the processor is responsible for: Communicating with memory, fetching instructions ahead of time, generating physical addresses, managing the instruction queue. Describe how this unit improves performance and explain its responsibilities. CLO3 3
(b) Which kinds of Register & Pointer work on Execution Unit? Identify their roles in program execution. CLO3 3
3. (a) Draw a simple block diagram of a reduced-instruction processor architecture and describe four major design features that give it performance advantages over traditional complex-instruction processors. CLO3 3
(b) Compare the older 8-bit architecture with the newer 16-bit architecture in terms of: CLO3 3
 - Addressing capability.
 - Instruction execution style.
 - Pipelining.
 - Size, I/O and Cost.
4. (a) In one addressing technique, operand address is stored in registers. Explain four categories this technique with numerical examples, showing how the final memory address is computed. CLO2 3
(b) Outline the major differences between Intel Core 2, Intel Core i7, Intel Core i5, Intel Core i3. CLO3 3
5. (a) Explain the following addressing mode with examples: CLO3 3
 - Immediate Addressing
 - Implied Addressing
 - Register Addressing

RISC may be

(b) The status indicator in 8086 includes

CLO3 3

- Bits that show arithmetic results
- Bits that influence program flow

Categorize these two groups, and explain how different operations set, reset, or use these bits.

Handwritten notes: A circle with an arrow pointing up and the letter 'I' below it, and another circle with an arrow pointing up and the letter 'O' below it.

6. (a) Draw the basic block diagram of a computer system. Explain the main Features of Microprocessor 8086. CLO3 3
- (b) Draw the pin diagram of Memory Banking and Examine the operation for different combination of enable keys (A_0 and BHE) in ODD and EVEN Bank. CLO3 3
7. (a) In 8086, instruction fetching and execution overlap. Explain the benefit of this mechanism with proper figure. CLO3 3
- (b) Explain with diagram how high-level code is transformed into machine-executable steps. Discuss the roles of: The program translation stage, The object code generation stage, The loading and execution stage. CLO1 3



HAMDARD UNIVERSITY BANGLADESH

Department of Computer Science and Engineering
Faculty of Science, Engineering and Technology

Course Code: MAT312
Midterm Examination
Date: 12/11/2025

Course Title: Engineering Mathematics
Semester: Fall 2025
Total Marks: 30

Total Time: 1:30 Hours

Answer any three (03) out of the following questions:

3 × 10 = 30

[The figures in the right margin indicate full marks. Symbols used have their usual meaning.]

No.	Questions	CLOs	Marks
Q1. (a)	Define Fourier series.	CLO1	[1]
(b)	Find the Fourier constants a_0 , a_n , and b_n within the interval $[-\pi, \pi]$.	CLO2	[5]
(c)	State and prove the Parseval's formula.	CLO2	[4]
Q2. (a)	Consider the following function: $f(x) = \begin{cases} -2, & \text{when } -3 < x < 0 \\ 2, & \text{when } 0 < x < 3 \end{cases}$		[6]
(i)	What is the period of the function?	CLO1	
(ii)	Find the Fourier coefficients corresponding to the function.	CLO3	
(iii)	Write the corresponding Fourier series.	CLO3	
(b)	Find the Fourier series of $f(x) = x$ in the interval $-\pi \leq x \leq \pi$.	CLO3	[4]
Q3. (a)	If $f(x) = \begin{cases} -1, & -\pi \leq x \leq 0 \\ 1, & 0 \leq x \leq \pi \end{cases}$, then show that $f(x) = \frac{4}{\pi} \left[\sin x + \frac{1}{3} \sin 3x + \frac{1}{5} \sin 5x + \dots \right]$.	CLO3	[6]
(b)	Find the series of sines and cosines which represents $x + x^2$ in the interval $-\pi \leq x \leq \pi$.	CLO3	[4]
Q4. (a)	Define Laplace and inverse Laplace transforms.	CLO1	[2]
(b)	Find the Laplace transform of the following functions: (i) e^{3t} and (ii) $\cos 5t$	CLO1	[4]
(c)	If $L\{F(t)\} = f(s)$, then show that (i) $L\{F'(t)\} = sf(s) - F(0)$ and (ii) $L\{F''(t)\} = s^2 f(s) - sF(0) - F'(0)$.	CLO2	[4]
Q5. (a)	Prove that the following: (i) $\int_0^{\infty} \frac{\sin at}{t} dt = \frac{\pi}{2}, a > 0$ and (ii) $L\left\{\frac{\cos at - \cos bt}{t}\right\} = \frac{1}{2} \ln\left(\frac{s^2 + b^2}{s^2 + a^2}\right)$	CLO3	[6]
(b)	Show that $L^{-1}\left\{\frac{6s-4}{s^2-4s+20}\right\} = 2e^{2t}(3 \cos 4t + \sin 4t)$.	CLO3	[4]

Handwritten notes and calculations:

$$f(-x) = -x$$

$$f(x) = x$$

$$f(x) = x$$

$$f(x) = x$$



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 317: Computer Networks

Patch: 25th

Semester: Fall 2025

Time: 1 Hour 30 Minutes

Date: 14/12/2025

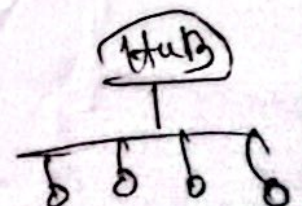
Mid-Term Examination

Total Marks: 30

Answer five questions of the following.

5 × 6 = 30

	CLOs	Marks
1(a). Briefly explain the basic characteristics of computer networks.	1	2
1(b). Explain the Star topology and the Mesh topology with advantages and disadvantages.	1	4
2(a). List the primary attributes of the Open Systems Interconnection (OSI) model.	1	2
2(b). Explain the functionalities of the Transport layer and the Lower layers of OSI model.	1	4
3(a). Define the Gateway and the Network Interface Card of computer networks.	4	2
3(b). Write the differences between HUB and Switch.	4	2
3(c). List the basic characteristics of layer 3 devices of TCP/IP model.	4	2
4(a). What are the Peer-to-Peer (P2P) network and the Client-Server network?	3	2
4(b). Write the comparisons among the Personal Area Networks, the Local Area Networks and the Campus Area Networks.	3	4
5(a). List the services of servers.	3	1
5(b). Define 'Token Ring'. How does the token ring pass messages in the computer networks?	5	2
5(c). Explain the Logical Link Control (LLC) of the Data link layer.	5	3
6(a). Consider IP address 192.168.10.0/25 for the given network. Find the first usable IP address of CSE lab, last usable IP address of EEE lab, 10th usable IP address of BBA lab and last usable IP address of link. The number of workstations for CSE, EEE and BBA labs are 25, 30 and 15.	5	3
6(b). A datagram of 5000 bytes (20 bytes of IP header) reached the router and must be forwarded to link with MTU (maximum transmission unit) of 800 bytes. How many fragments will be granted? Also, write MF, offset and total length values for all.	2	3



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|-------|--|---|---|
| 7(a). | How many classes are in Internet Protocol? Write the names, ranges and subnet mask of each class. | 4 | 2 |
| 7(b). | Illustrate the datagram header of IPv4. Briefly explain the flag bits and fragment offset of IPv4. | 4 | 4 |



Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE311: Theory of Computing

Batch: 25th

Semester: Fall 2025

Date: 07/12/2025

Mid-Term Examination

Time: 1 Hour 30 minutes

Total Marks: 30

Answer any five questions.

Marks: 5×6 = 30

			Marks	CLOS
1.	(a)	Write the uses of automata.	2	1
	(b)	Explain the five key terms in defining a DFA.	4	2
2.	(a)	Compare between Σ^+ and Σ^* , when $\Sigma = \{abc\}$.	2	2
	(b)	Draw an NFA for the string $a^+ + (ab)^*$, over the alphabet $\Sigma = \{a,b\}$.	4	3
3.	(a)	With a proper diagram, differentiate between NFA and DFA.	3	2
	(b)	Write short notes on i) Alphabet ii) Power of Alphabet iii) Concatenation of string	3×1=3	2
4.	a)	Define a Moore Machine and mention its tuples.	2	2
	b)	Design a Moore machine for a binary input sequence such that if it has a substring 1011, the machine outputs A) if the input has substring 1101, it outputs B) otherwise it outputs C.	4	3
5.	Construct an equivalent DFA from the NFA below:		6	3
<pre> graph LR 1((1)) -- a --> 2((2)) 1 -- b --> 3((3)) 2 -- a --> 1 2 -- a --> 4((4)) 3 -- a --> 2 3 -- b --> 4 4 -- b --> 1 4 -- "a,b" --> 4 style 1 fill:#fff,stroke:#000,stroke-width:2px style 2 fill:#fff,stroke:#000,stroke-width:2px style 3 fill:#fff,stroke:#000,stroke-width:2px style 4 fill:#fff,stroke:#000,stroke-width:2px </pre>				
6.	Design a DFA accepting the set of all strings starts with 'abc', over the alphabet $\{a,b,c\}$. Show that abcabc is accepted by the DFA.		6	3
7.	Suppose you are given the alphabet $\Sigma = \{0,1\}$. Draw a DFA that accepts the following strings: i) Even number of 0's. ii) Even number of 1's. iii) Even number of 0's and 1's.		6	3



HAMMARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 313 Database Management System

Batch: 25th

Date: 17/12/2025

Mid Term Examination

Semester: Fall 2025

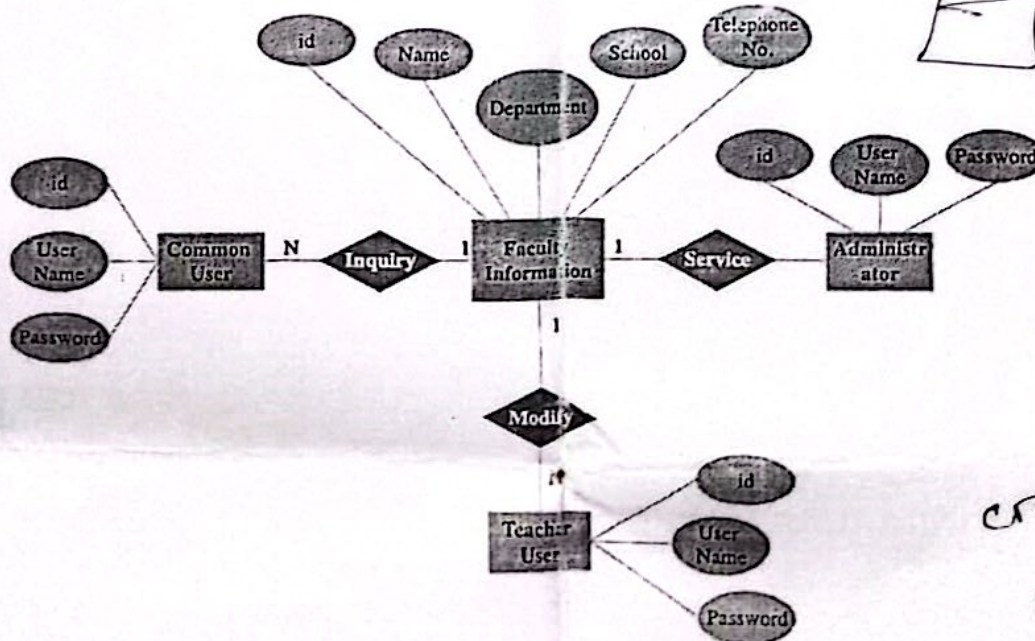
Time: 1 hour 30 minutes

Total Marks: 30

Answer any five questions from the following.

5 x 6 = 30

1. (a) Write the steps of feasibility analysis of a Database Management System (DBMS). CLO 1 2
(b) Briefly explain different types of feasibility analysis of a DBMS. CLO 1 4
2. (a) Define context diagram with an example. CLO 1 2
(b) State the art, arrangement and guidelines for a successful interview to determine DBMS requirements. CLO 1 4
3. (a) How one can define Relational DBMS? CLO 3 2
(b) State the advantages of DBMS. CLO 1 2
(c) State the responsibilities of a database administrator. CLO 1 2
4. (a) Define the degrees of relationship of a DBMS. CLO 1 3
(b) Explain the steps of constructing an E-R diagram. CLO 1 3
5. (a) Define Data Flow Diagram (DFD). CLO 1 2
(b) Briefly explain the questionnaires of a DBMS with its advantages. CLO 1 4
6. Write the DFD rules of a Database Management System. CLO 1 6
7. Consider the following E-R diagram: CLO 2 6



Write the SQL command to create all the relations. Find the Questionnaires for the given E-R diagram.